

AIG130 – Lab 3

Objective

Apply the concepts learned in the lecture to a real-world machine learning engineering problem.

Group Formation

Students can work individually or form groups of up to 4 members. Collaboration and effective communication are encouraged for those working in groups.

Assignment

Select a scenario that represents a common challenge in the field of ML Engineering. Your task will be to apply the concepts learned to propose a comprehensive solution. A sample scenario will be provided below as a guide.

Scenario Template

Imagine you are part of a machine learning engineering team at a <YOUR_CHOICE> company. Your team's goal is to develop a machine learning model to <YOUR_CHOICE>. The model will help the company to <YOUR_CHOICE>.

Suggested Steps

1. Case Selection:
 - a. You can choose a case from a range of sectors such as healthcare, finance, retail, or technology.
 - b. The case should involve implementation of ML Engineering principles. You may preferably choose to use any cloud service for machine learning, but it is not mandatory for this lab.
2. ML Engineering Pipeline would include:
 - a. Data Collection and Cleaning:
 - i. Describe the types of data you would collect for this model.
 - ii. Explain how you would clean and preprocess this data.
 - b. Model Development and Training (THIS STEP IS OPTIONAL AS WE HAVE NOT EXPLORED ML ARCHITECTURES):
 - i. Choose a suitable machine learning algorithm for this problem.
 - ii. Explain your choice and how you would train this model with the collected data.
 - c. Validation and Testing:
 - i. Discuss the methods you would use to validate and test the model's accuracy and effectiveness.
 - d. Deployment and Monitoring:

- i. Propose a strategy for deploying this model.
 - ii. Outline how you would monitor and update the model once it is deployed.
- e. Collaboration:
 - i. Describe how you would collaborate with company staff/departments throughout this project.
- f. Addressing Challenges:
 - i. Identify potential challenges you might face (e.g., data privacy, model drift) and how you would address them.
- g. MLOps:
 - i. Explain how MLOps principles can be applied to streamline deployment, monitoring, and maintenance of your ML model.

Deliverables

Report: Submit a PDF report covering all three tasks. Include:

- An introduction and conclusion.
- Sections for each task with headings.
- Visual elements (charts, tables) where applicable.

Presentation: Prepare and deliver a presentation of your findings in class. The presentation is also required to be submitted in Blackboard. It can be as PDF, PPTX, Link, etc.

- Duration: Up to 15 minutes per group.
- Cover key points and findings from the report.
- Encourage questions and discussions.

Individual Reflection

Each student must submit a separate reflection document answering the following:

1. If you worked individually, why did you choose to go solo? What challenges did you encounter?
2. If you did it as a group, how were tasks distributed, and what was your specific task? What challenges did you encounter?
3. Why do you think ML Engineering is important for effective implementation of ML projects? There is no right or wrong answer. Explain and elaborate on the answer that you give.

Submission

Submissions are done through the provided links in Blackboard. There are three links:

1. Group report (one submission per group): 4%
2. Individual reflection: 2%
3. Presentation file/link: 2%

Assessment Criteria

- Quality and depth of research and analysis.
- Clarity and organization of the report.

- Effectiveness of the presentation.
- Thoughtfulness of the individual reflection.